



EE202 – Object Oriented Programming Project

Instructor: Ayman Madbouly

Team#D

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1. Project Overview:

This project idea is designing and developing a scientific calculator application using Python and the PyQt framework. The main purpose of the project is to apply Object-Oriented Programming (OOP) concepts to create a well-organized and user-friendly desktop application.

The application consists of three main components: a scientific calculator, an equation solver, and a function plotting module. The scientific calculator is designed to perform basic and advanced mathematical operations, including trigonometric functions, logarithms, and powers. The equation solver is intended to solve two linear equations with two unknowns, while the function plotting module allows users to visualize mathematical functions using Matplotlib.

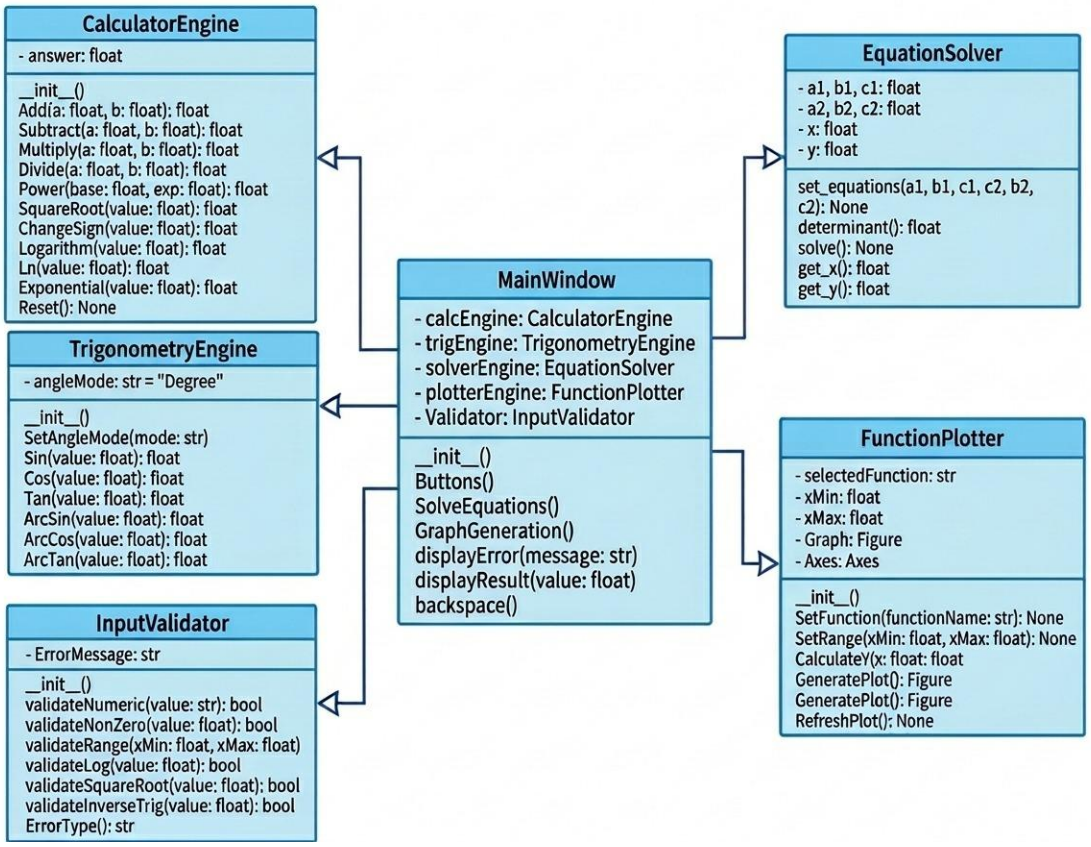
To achieve these objectives, the application is divided into several classes, each with a specific responsibility. This approach helps organize the code, separate the graphical user interface from the calculation logic, and simplify testing and future modifications. Through this project, the team aims to improve its practical understanding of OOP, GUI development, and software integration.

2. Team D - Members & Responsibilities:

This Table will display the members of Team D and contribution of each for each checkpoint of our EE202 Project. This table will help us to make sure that we divided the work fairly and didn't give any member more work than the others.

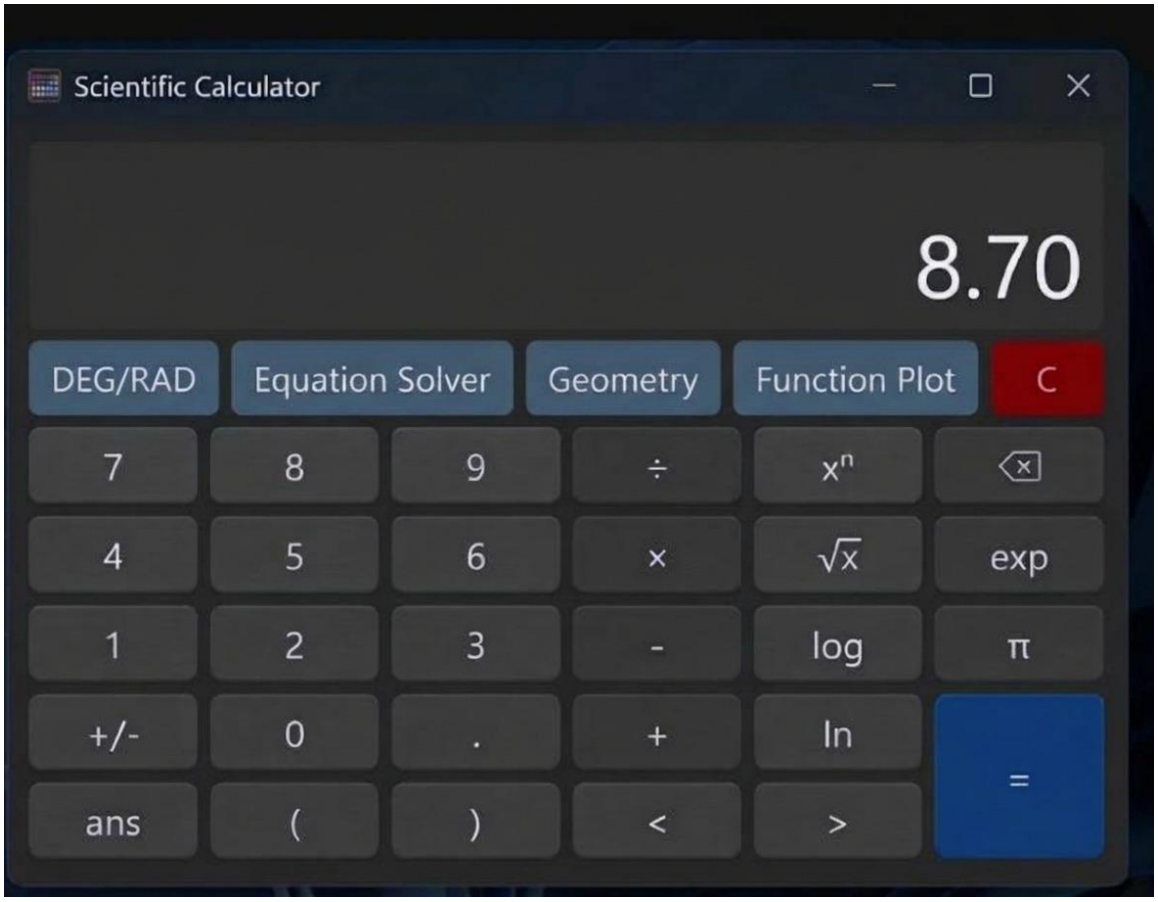
Team Members	Checkpoint 1	Checkpoint 2	Checkpoint 3	Checkpoint 4	Checkpoint 5
Abdulrahman	Team member responsibilities and FunctionPlotter UML	Degree/Rad mode, implementation of FunctionPlotter Class.	Required plotting functions and user-controlled plotting intervals, Updated FunctionPlotter UML.	Final UML and code of Function Plotter, Final technical report.	Complete Function Plotter and Final documentation.
Mohammed	MainWindow UML and Calculator Engine UML.	Basic calc functions, implementation of Calculator Engine and Main Window classes.	Updated Main Window and Calculator Engine UMLs.	Final UML and code of MainWindow and Calculator Engine	Complete Main Window and Calculator Engine.
Abdulelah	Equation Solver UML and Trigonometry Engine UML.	Scientific calc functions, implementation of EquationSolver and TrigonometryEngine classes.	Working twoequation solver, Updated EquationSolver and TrigonometryEngin e UMLs.	Final UML and code of EquationSolver and Trigonometry Engine.	Complete EquationSolver and TrigonometryEngine.
Eyad	PyQt GUI and InputValidator UML.	Working PyQt interface, implementation of InputValidator class.	Updated InputValidator UML, improvement error handling and initial testing.	Final UML and code of InputValidator, Final presentation slides.	Complete InputValidator and AI usage record.

3. Initial UML Class Diagram



Description: The UML Class Diagram illustrates the object-oriented architecture of the application. It highlights the separation between the MainWindow (the Graphical User Interface) and the backend calculation logic represented by the various engine classes (CalculatorEngine, TrigonometryEngine, EquationSolver, FunctionPlotter, and InputValidator). The MainWindow holds instances of these engines, delegating mathematical tasks to ensure a modular and maintainable codebase.

4. Graphical User Interface (GUI) Design (initial)



Description: The initial graphical user interface design provides a clear, intuitive layout. It features a top navigation bar allowing users to switch seamlessly between the Scientific Calculator, Equation Solver, and Function Plotting modules without overcrowding the screen. The main view utilizes a standard grid layout for numbers and operations, maintaining a professional appearance suitable for scientific applications.